IND3001 Engineering Economics

Asst. Prof. Çağın Erbek

caginerbek@haliç.edu.tr Office:8331

ELEMENTARY ECONOMIC ANALYSIS

- Whether it is a business situation or a day-to-day event in somebody's personal life, there are a large number of economic decision making involved.
- One can manage many of these decision problems by using simple economic analysis.
- For example, an industry can source its raw materials from a nearby place or from a far-off place.
- In this problem, the following factors will affect the decision:

· Price of the raw material

· Transportation cost of the raw material

· Availability of the raw material

· Quality of the raw material

ELEMENTARY ECONOMIC ANALYSIS

Consider the alternative of sourcing raw materials from a nearby place with the following characteristics:

- · The raw material is more costly in the nearby area.
- · The availability of the raw material is not sufficient enough to support the operation of the industry throughout the year.
- · The raw material requires pre-processing before it is used in the production process. This would certainly add cost to the product.
- · The cost of transportation is minimal under this alternative.

On the other hand, consider another alternative of sourcing the raw materials from a far-off place with the following characteristics:

- · The raw material is less costly at the far off place.
- · The cost of transportation is very high.
- · The availability of the raw material at this site is abundant and it can support the plant throughout the year.
- · The raw material from this site does not require any pre-processing before using it for production.

Under such a situation, the procurement of the raw material should be decided in such a way that the overall cost is minimized.

The concept of simple economic analysis is illustrated using suitable examples in the following areas

·Material selection for a product

· Design selection for a product

· Design selection for a process industry

· Building material selection for construction activities

· Process planning/Process modification

Material Selection for a Product/ Substitution of Raw Material

- The cost of a product can be reduced greatly by substitution of the raw materials. Among various elements of cost, raw material cost is most significant and it forms a major portion of the total cost of any product.
- So, any attempt to find a suitable raw material will bring a reduction in the total cost in any one or combinations of the following ways:

- · Cheaper raw material price
- · Reduced machining/process time
- · Enhanced durability of the product

EXAMPLE -1

In the design of a jet engine part, the designer has a choice of specifying either an aluminium alloy casting or a steel casting. Either material will provide equal service, but the aluminium casting will weigh 1.2 kg as compared with 1.35 kg for the steel casting.

The aluminium can be cast for $80.00 per kg. and the steel one for 35.00 $ per kg.

The cost of machining per unit is $150.00 for aluminium and $170.00 for steel.

Every kilogram of excess steel weight is associated with a penalty of $1,300 due to increased fuel consumption.

Which material should be specified and what is the economic advantage of the selection per unit?

Design Selection for a Product

The design modification of a product may result in reduced raw material requirements, increased machinability of the materials and reduced labour.

Design is an important factor which decides the cost of the product for a specified level of performance of that product.

EXAMPLE -2

Two alternatives are under consideration for a tapered fastening pin. Either design will serve the purpose and will involve the same material and manufacturing cost except for the lathe and grinder operations.

Design A will require 16 hours of lathe time and 4.5 hours of grinder time per 1,000 units.

Design B will require 7 hours of lathe time and 12 hours of grinder time per 1,000 units.

The operating cost of the lathe including labour is Rs. 200 per hour.
The operating cost of the grinder including labour is Rs. 150 per hour.

Which design should be adopted if 100,000 units are required per year and what is the economic advantage of the best alternative?

Building Material Selection

The sourcing of raw materials will have a significant effect on the cost of any product. Hence, it is assumed that the price of raw material is location dependent.

While sourcing a raw material, the cost of transportation is to be considered in conjunction with the price of the raw material.

EXAMPLE -3

In the design of buildings to be constructed in Alpha State, the designer is considering the type of window frame to specify. Either steel or aluminium window frames will satisfy the design criteria.

Because of the remote location of the building site and lack of building materials in Alpha State, the window frames will be purchased in Beta State and transported for a distance of 2,500 km to the site.

The price of window frames of the type required is Rs. 1,000 each for steel frames and Rs. 1,500 each for aluminium frames.

The weight of steel window frames is 75 kg each and that of aluminium window frame is 28 kg each.

The shipping rate is Re 1 per kg per 100 km.

Which design should be specified and what is the economic advantage of the selection?

Process Planning /Process Modification

While planning for a new component, a feasible sequence of operations with the least cost of processing is to be considered. The process sequence of a component which has been planned in the past is not static. It is always subject to modification with a view to minimize the cost of manufacturing the component.

So, the objective of process planning/process modification is to identify the most economical sequence of operations to produce a component.

The steps in process planning are as follows:
1. Analyze the part drawing to get an overall picture of what is required.
2. Make recommendations to or consult with product engineers on product design changes.
3. List the basic operations required to produce the part to the drawing or specifications.
4. Determine the most practical and economical manufacturing method and the form or tooling required for each operation.
5. Devise the best way to combine the operations and put them in sequence.
6. Specify the gauging required for the process.

EXAMPLE -4

The process planning engineer of a firm listed the sequences of operations as shown in Table to produce a component.

Sequence	Process sequence			
1	Turning	Milling	Shaping	Drilling
2	Turning	Milling	Drilling	
3	All operations are performed with CNC machine			

The details of processing times of the component for various operations and their machine hour rates are summarized in Table below;

Operation	Machine Hour Rate (Rs)	Process Sequence (minutes)		
		1	2	3
Turning	200	5	5	-
Milling	400	8	14	-
Shaping	350	10	-	-
Drilling	300	3	3	-
CNC Operations	1000	-	-	8

Time Value of Money



Thanks!
Q&A

